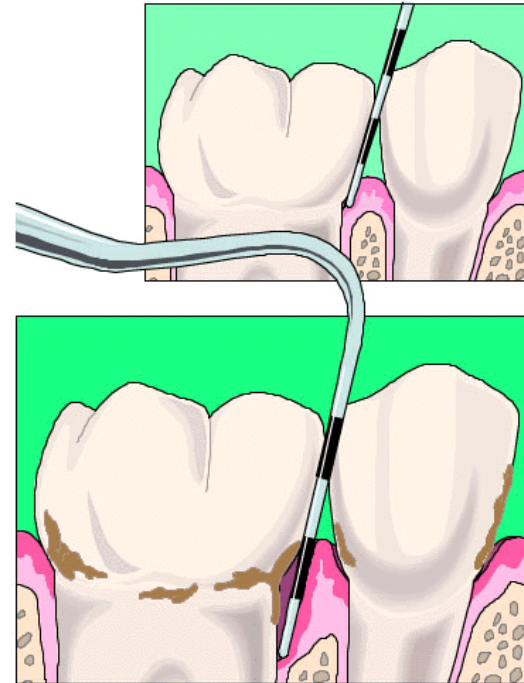


Dilemma at the Dentist's



What is the probability of a cavity given a toothache?
What is the probability of a cavity given the probe catches?

Inference by Enumeration

Start with the joint distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>			.072	.008
$\neg$ <i>cavity</i>			.144	.576

For any proposition ϕ , sum the atomic events where it is true:

$$P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$$

$$\begin{aligned} P(\text{toothache}) &= .108 + .012 + .016 + .064 \\ &= .20 \text{ or } 20\% \end{aligned}$$

Inference by Enumeration

Start with the joint distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>				
$\neg$ <i>cavity</i>			.144	.576

For any proposition ϕ , sum the atomic events where it is true:

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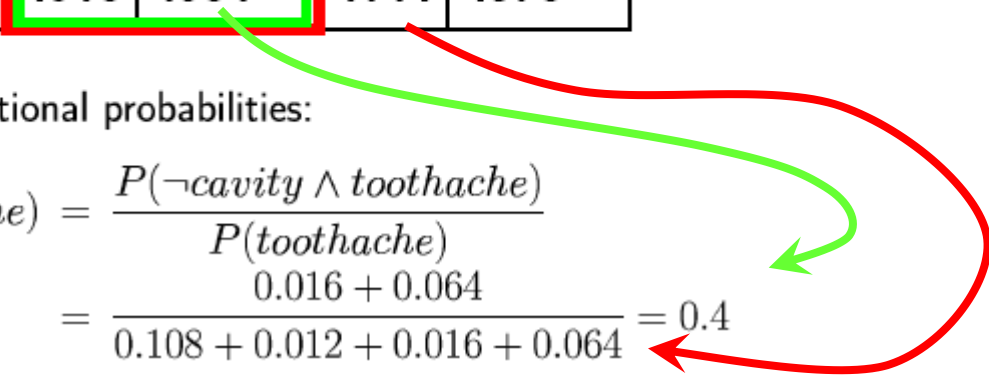
$$P(\text{toothache} \vee \text{cavity}) = .20 + .072 + .008$$
$$.28$$

Inference by Enumeration

Start with the joint distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

Can also compute conditional probabilities:

$$\begin{aligned} P(\neg \text{cavity} | \text{toothache}) &= \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})} \\ &= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4 \end{aligned}$$


Complexity of Enumeration

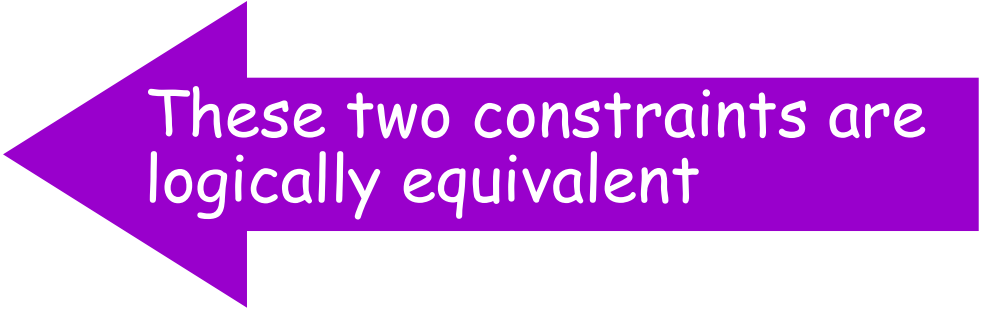
- Worst case time: $O(d^n)$
 - Where d = max arity
 - And n = number of random variables
- Space complexity also $O(d^n)$
 - Size of joint distribution
- Prohibitive!

Independence

- A and B are *independent* iff:

$$P(A \mid B) = P(A)$$

$$P(B \mid A) = P(B)$$



These two constraints are logically equivalent

- Therefore, if A and B are independent:

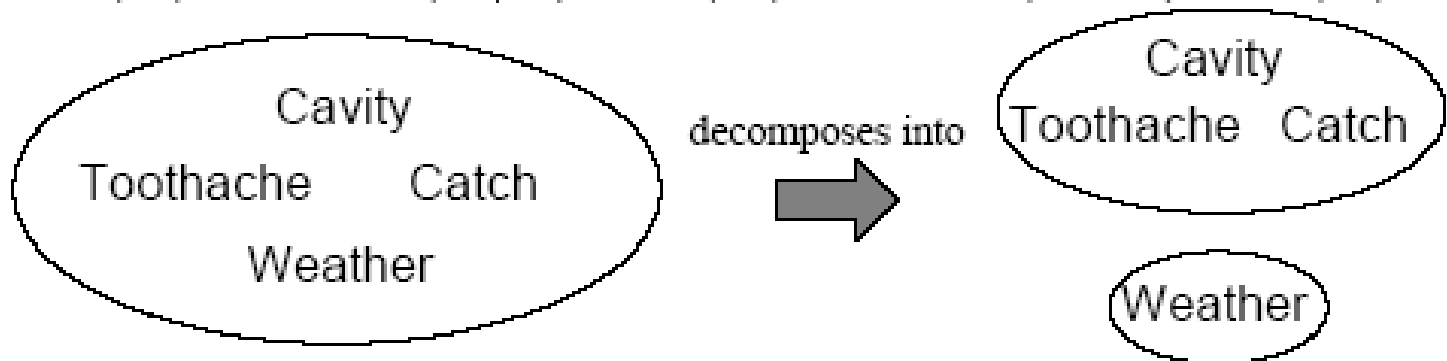
$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)} = P(A)$$

$$P(A \wedge B) = P(A)P(B)$$

Independence

A and B are independent iff

$$\mathbf{P}(A|B) = \mathbf{P}(A) \quad \text{or} \quad \mathbf{P}(B|A) = \mathbf{P}(B) \quad \text{or} \quad \mathbf{P}(A, B) = \mathbf{P}(A)\mathbf{P}(B)$$



$$\begin{aligned} &\mathbf{P}(\textit{Toothache}, \textit{Catch}, \textit{Cavity}, \textit{Weather}) \\ &= \mathbf{P}(\textit{Toothache}, \textit{Catch}, \textit{Cavity})\mathbf{P}(\textit{Weather}) \end{aligned}$$

32 entries reduced to 12; for n independent biased coins, $2^n \rightarrow n$

Complete independence is powerful but rare
What to do if it doesn't hold?

Conditional Independence

$P(\textit{Toothache}, \textit{Cavity}, \textit{Catch})$ has $2^3 - 1 = 7$ independent entries

If I have a cavity, the probability that the probe catches in it doesn't depend on whether I have a toothache:

$$(1) P(\textit{catch}|\textit{toothache}, \textit{cavity}) =$$

The same independence holds if I haven't got a cavity:

$$(2) P(\textit{catch}|\textit{toothache}, \neg \textit{cavity}) = P(\textit{catch}|\neg \textit{cavity})$$

Conditional Independence II

$$P(\text{catch} \mid \text{toothache}, \text{cavity}) = P(\text{catch} \mid \text{cavity})$$

$$P(\text{catch} \mid \text{toothache}, \neg \text{cavity}) = P(\text{catch} \mid \neg \text{cavity})$$

Equivalent statements:

$$\mathbf{P}(\text{Toothache} \mid \text{Catch}, \text{Cavity}) = \mathbf{P}(\text{Toothache} \mid \text{Cavity})$$

$$\mathbf{P}(\text{Toothache}, \text{Catch} \mid \text{Cavity}) = \mathbf{P}(\text{Toothache} \mid \text{Cavity}) \mathbf{P}(\text{Catch} \mid \text{Cavity})$$

Why only 5 entries in table?

Write out full joint distribution using chain rule:

$$\mathbf{P}(\text{Toothache}, \text{Catch}, \text{Cavity})$$

$$= P(\text{Cavity})P(\text{Catch} \mid \text{Cavity})P(\text{Toothache} \mid \text{Catch}, \text{Cavity})$$

$$= P(\text{Cavity})P(\text{Catch} \mid \text{Cavity})P(\text{Toothache} \mid \text{Cavity})$$

$$= 1 + 2 + 2 = 5 \text{ params are needed}$$

Power of Cond. Independence

- Often, using conditional independence reduces the storage complexity of the joint distribution from exponential to linear!!
- Conditional independence is the most basic & robust form of knowledge about uncertain environments.

Bayes Rule

Bayes rules!



posterior

$$P(x, y) = P(x | y)P(y) = P(y | x)P(x)$$

$$P(x | y) = \frac{P(y | x) P(x)}{P(y)} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

Same for all values of x
Hence we won't consider this mostly

Useful for assessing **diagnostic** probability from **causal** probability:

$$P(Cause | Effect) = \frac{P(Effect | Cause)P(Cause)}{P(Effect)}$$

Computing Diagnostic Prob. from Causal Prob.

$$P(Cause|Effect) = \frac{P(Effect|Cause)P(Cause)}{P(Effect)}$$

E.g. let M be meningitis, S be stiff neck

$$P(M) = 0.0001,$$

$$P(S) = 0.1,$$

$$P(S|M) = 0.8$$

$$P(M|S) = \frac{P(s|m)P(m)}{P(s)} = \frac{0.8 \times 0.0001}{0.1} = 0.0008$$

Note: posterior probability of meningitis still very small!

Other forms of Bayes Rule

$$P(x|y) = \frac{P(y|x) P(x)}{P(y)} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

$$P(x|y) = \frac{P(y|x) P(x)}{\sum_x P(y|x) P(x)}$$

$$P(x|y) = \alpha P(y|x) P(x)$$

posterior $\propto$ likelihood $\cdot$ prior

to remove proportionality add a constant here
it is Alpha

therefore make it proportional

Conditional Bayes Rule

If conditioned by z as w
then condition with z
everywhere even in the
rhs

$$P(x \mid y, z) = \frac{P(y \mid x, z) P(x \mid z)}{P(y \mid z)}$$

no need to add z in the summation

$$P(x \mid y, z) = \frac{P(y \mid x, z) P(x, z)}{\sum_x P(y \mid x, z) P(x \mid z)}$$

$$P(x \mid y, z) = \alpha P(y \mid x, z) P(x \mid z)$$

Bayes' Rule & Cond. Independence

$$\begin{aligned} & \mathbf{P}(Cavity|toothache \wedge catch) \\ &= \alpha \mathbf{P}(toothache \wedge catch|Cavity)\mathbf{P}(Cavity) \\ &= \alpha \mathbf{P}(toothache|Cavity)\mathbf{P}(catch|Cavity)\mathbf{P}(Cavity) \end{aligned}$$

This is an example of a *naive Bayes* model:

$$\mathbf{P}(Cause, Effect_1, \dots, Effect_n) = \mathbf{P}(Cause) \prod_i \mathbf{P}(Effect_i|Cause)$$



Total number of parameters is *linear* in n